YIFEI SUN

Staff Software Engineer · Identity Platform, API Architecture & 0→1 Product Builds · 13 Years

(617) 922-4823
sun1fei@gmail.com
linkedin.com/in/yifei-sun-a789861b

SUMMARY

Staff-level platform engineer with 13 years building identity infrastructure, developer-facing APIs, and 0→1 product systems at scale. Architected and shipped a production identity platform — Keycloak-based SSO with OAuth 2.0 and SAML federation, custom token validation and RBAC enforcement at API gateway, and zero-trust service mesh — unifying auth across internal systems, enterprise partners, and external consumers. Led a multi-quarter API platform redesign serving millions of daily requests (p99 250ms→50ms, 99.9% availability), driving architecture and API design standards across 10+ teams. Currently building AI-powered developer tooling (MCP server on OpenAI Apps SDK, LLM-based automated validation) and using AI assistants extensively in daily engineering workflow. Combines deep platform engineering with cross-organizational technical leadership, strong API design judgment, and multiple 0→1 product deliveries from blank page to production.

LEADERSHIP HIGHLIGHTS

- Built production identity platform: Keycloak SSO, OAuth 2.0/SAML federation, RBAC enforcement at gateway, zero-trust service mesh (mTLS)
- Delivered multiple 0→1 products: identity platform, AI developer tools (MCP server), digital commerce platform, GraphQL federation layer
- Led multi-quarter API platform redesign (8K rpm, millions daily): defined API contracts for multiple consumer types, backward-compatible migrations
- Building and using AI tools daily: developed MCP server and LLM-based validation systems; extensive use of AI-assisted development in production workflows

Identity & Security: OAuth 2.0, SAML, OIDC, Keycloak, JWT/token validation, RBAC/ABAC, mTLS, Kong, Nginx · **Platform & APIs:** Java (13 yr), Python, Node.js/TypeScript, GraphQL (Netflix DGS), REST, gRPC, Protobuf · **Infrastructure:** AWS (ECS, EKS, IAM, KMS, SQS, SNS), Kubernetes, Kafka, Redis, PostgreSQL, Terraform · **AI/LLM:** OpenAI SDK, MCP Protocol, Claude Code, LLM-based automation & detection tooling · **Observability:** Datadog, Grafana, Distributed Tracing

EXPERIENCE

CarGurus — Principal Software Engineer / Technical Lead

Oct 2019 – Present

AI Developer Tools — MCP Server & LLM-Based Validation (0→1)

- Sole backend engineer designing and shipping an MCP server on the OpenAI Apps SDK — defined API contracts, tool schemas, access control boundaries, and request/response interfaces for external AI consumer integration. Delivered under ambiguity, iterating rapidly with PMs and directors on API surface design.
- Designing AI-powered cross-domain validation: LLM-based impact analysis + deterministic spec validation to detect breaking changes across service boundaries at PR time. Architecture: rule-based pre-filter cascading to semantic analysis, consumer registry tracking field-level dependencies.
- Extensive daily use of AI-assisted development tools (Claude Code) for production engineering — code generation, system design iteration, and automated code review workflows.

High-Traffic API Platform — Multi-Consumer Architecture

- Led 3 principal/senior engineers and aligned 10+ teams across 3 departments to redesign the core API platform (~8,000 rpm, millions of daily impressions), defining API contracts, backward-compatible migration strategy, and safe rollout across multiple consumer types (frontend, backend, AI, mobile).
- Architected orchestration layer consolidating 12 downstream integrations into 5 services with well-defined boundaries, per-service auth, circuit breaking, and fault isolation. Built automated parity testing comparing new vs. legacy responses at 50% production traffic.

  p99: 250 ms → 50 ms (5×) **Availability: 98% → 99.9%** **Deploy: 1 hr → 3 min** (20×)

GraphQL Federation — Unified API Layer for Multiple Consumer Types

- Designed Netflix DGS subgraph for federated GraphQL gateway — unified, access-controlled API serving frontend, backend, and AI consumers through a single graph with per-consumer authorization and consistent contracts.
- Implemented schema federation, batch data loaders (N+1 prevention), and JVM warmup strategies; built load and integration tests ensuring backward compatibility across all consumer types.

Digital Commerce Platform (0→1) & Experimentation

- Led backend design and launch of a new digital commerce product from blank page to production during the pandemic — microservice and event-driven architecture for inventory, merchandising, and lead capture flows.
- Designed for safe experimentation: feature flags, isolated experiment paths, and clean retirement without destabilizing core systems. Multiple product variants launched and retired using this platform.

Cross-Org Architecture Standards & Engineering Excellence

- Established reusable patterns for API design, caching, and distributed orchestration adopted across multiple teams; authored design docs, RFCs, and architecture decision records driving cross-team consensus.
- Mentored senior engineers on API craft, system design, and path to Principal-level scope. Led design reviews for major platform changes.

Applause — Senior Software Engineer / Technical Lead

Oct 2017 – Oct 2019

Identity Platform — SSO, Federation & API Gateway Auth (0→1)

- Architected and deployed a production identity platform: migrated legacy auth to Keycloak-based SSO with OAuth 2.0 and SAML identity federation, unifying authentication across internal systems, an enterprise partner platform, and a large external community with fine-grained RBAC.
- Built custom token validation and RBAC enforcement plugins for Kong API gateway (Python) — per-request JWT validation, token refresh handling, session management, and permission checks at the gateway layer before requests reach backend services.
- Deployed service mesh (Linkerd on Kubernetes) with mTLS for zero-trust service-to-service communication. Coordinated migration of multiple Java and Node.js services behind the new auth gateway with minimal downtime, aligning 4 teams on consistent identity posture.

Event Sourcing & Platform Architecture

- Drove design of microservice architecture with event sourcing and CQRS; defined service isolation patterns, data boundaries, and access control models adopted across new domains.

Toast Inc. — Senior Software Engineer (Early Engineer)

Sep 2014 – Oct 2017

Core Platform Services & API Infrastructure (0→1)

- Among the earliest engineers during hypergrowth; built core platform services from scratch (Order Service, Labor Service, REST APIs) with authentication, authorization, and input validation at every boundary — systems that grew to process billions in GMV.
- Built custom API gateway (Nginx + Lua) for rate limiting, request authentication, traffic routing, and abuse prevention — the first externally-facing API security layer, built before dedicated security or DevOps teams existed.
- Set up initial AWS infrastructure with Ansible — provisioned environments, deployment pipelines, and IAM policies. Established foundational platform patterns during rapid 0→1 scaling.

Third-Party API Integrations & Developer Experience

- Led authenticated bidirectional API integrations with partners — schema validation, backward-compatible versioning, access control, and clear developer documentation for external consumers.

The MathWorks Inc. — Software Engineer II

Jul 2013 – Aug 2014

Developed internal engineering tools (Java) for MATLAB/Simulink workflows. Improved build/test automation and developer productivity tooling.

EDUCATION

M.S. in Electrical and Computer Engineering — Northeastern University, Boston 2010 – 2013

B.S. in Information Science — Beijing University of Aeronautics and Astronautics 2006 – 2010